

Import Price Shocks, FX-Monetary Policies, and Real Income Stabilization

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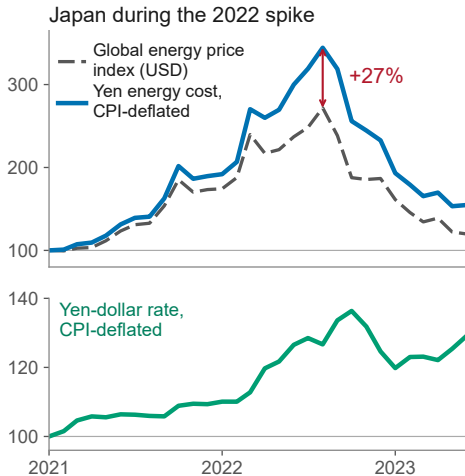
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Essential Import Price Spikes

- Recent years: **import-dependent economies** facing recurrent episodes of large, temporary spikes in essential-import prices:
 - 2022: LNG and crude +100-500%
 - 2026 Q1: crude +60%; Asian LNG +140%
- For an import-price-taking economy, this is a **negative terms-of-trade** shock: give away more domestic output to buy the import. Price-inelastic demand: shock translates into **real-income losses**.
- In the data: concurrent **real depreciation** endogenously worsens the terms of trade and amplifies real-income losses.
- Policy concerns in a monetary economy: **imported inflation** and nominal depreciation.

How should monetary and exchange rate policies respond?



Existing Policy Views

"It is far simpler to allow one price to change, namely, the price of foreign exchange. . . "

Friedman (1953, p. 173)

- **Textbook open-economy RA NK framework:**

- **"Raise rates:"** use monetary policy to anchor inflation; contractionary stance that limits aggregate demand and domestic activity.
- **Exchange rate as a shock absorber:** do not manage the real exchange rate. **Let it depreciate.**

Rationale: depreciation reflects expenditure switching. With perfect asset-market access, a higher import bill only a PDV loss. Households optimally smooth over time.

- **HANK earnings real income channel:** monetary policy less contractionary than RA, activity expansion indirectly supports real incomes of borrowing-constrained households.

The Policy Set

Monetary policy

- Operates through nominal rigidities
- Influences activity: output, employment
- Money growth, nominal rate

FX policy

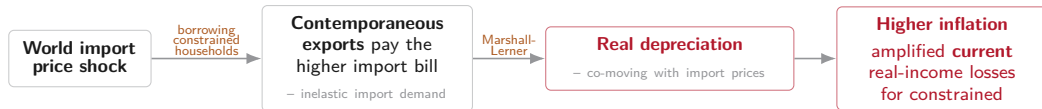
- Distorts international financial flows
- Influences real exchange rate path
- Capital control taxes, FX interventions

- Typically: FX policy for managing volatile capital flows, financial crises, macroprudential policy, and ToT manipulation.
- Monetary policy for fighting inflation and **protecting real incomes** of constrained households.

This Paper

- Open economy framework.
 - Import-dependent economy, small in import and asset markets. Large in export markets.
 - **Marshall-Lerner condition**: Inelastic import demand. Downward-sloping, price-elastic export demand: **real depreciation** required for higher export revenues.
 - **Heterogeneous households**: some borrowing constrained—cannot smooth consumption through the import price shock.
 - **No direct fiscal transfers** to constrained households.

From shock absorber to inflation amplifier:



How does heterogeneity change the roles of monetary and FX policies? To what extent can they protect households who cannot borrow through the shock?

Main Result: FX Policy and Real Income Stabilization

- A laissez-faire “let it depreciate” exchange rate response is costly for constrained households.
- Instead, policy *should lean against a contemporaneous depreciation* to **protect their real incomes** during shock episodes.
- **Active role for FX policy** under household heterogeneity in managing the real exchange rate path during episodes of temporary spikes in essential-import prices and associated inflation.

FX policy and indirect real income effects:



Main Result: FX Policy and Real Income Stabilization

- **The real-income stabilization channel of exchange rate policy:**
 - Induces stronger capital inflows during a spike and makes foreign purchasing power (“dollars”) more abundant in the FX market.
 - Finances more of the current import bill through capital inflows and less through current exports: retains more output at home.
 - Relies on the **Marshall-Lerner condition** to appreciate and **indirectly** boost real incomes.
 - Unwinds through capital outflows once the spike has passed: depreciate and export more; re-times the loss, does not mitigate it in PDV terms.
 - **Stabilized real income and consumption for constrained households**; stronger capital inflow and weaker export and RER co-movement with import prices: **stabilized terms of trade**.
- **FX policy as an anti-inflation policy:** more effective than monetary policy in protecting real incomes of constrained households and fighting imported inflation.
- Works by driving **interest parity wedges**. Monetary policy cannot. The interest parity condition limits any real income protection that monetary policy can deliver in isolation.

Results and Contributions

- **Normative analysis.** Jointly optimal monetary and FX policy response to a ToT shock.
 - **FX policy leads with real-income protection and disinflation.** Indirect income effects that rely on the Marshall-Lerner condition. Role increasing in the essentiality of imports, share of constrained.
 - **Monetary policy:** weaker role under final-good imports; employment-import complementarity: role increasing and complementary to FX policy. **More contractionary than RA due to a redistributive price effect:** import-induced depreciations hurt workers and benefit borrowers.
- **Quantitative two-agent model.** Nominal wage rigidity, financial intermediation frictions. Utilitarian-optimal FX interventions and monetary policy under perfect-foresight transitions.
Application: energy-price shocks. Japan calibration, 2022-like price path:
 - **Joint optimum vs conventional:** energy 19.49% cheaper; 29.03% of real wage loss undone. Average inflation: 0.60% → 0.23%
 - **Monetary policy alone:** 0.27% of wage loss undone; average inflation: 0.60% → 0.58%
- **HA framework with idiosyncratic risk and incomplete markets:** endogenously evolving constrained mass; jointly utilitarian-optimal perfect-foresight policies under timeless commitment.
 - Resolve a distributional time-inconsistency problem.
 - Computationally tractable nonlinear solution: exploit the affine form of the promise-keeping state.
 - Analyze the impact of FX policy's indirect income effects on consumption distribution.

- **Conventional open-economy policy under ToT shocks.** Use contractionary monetary policy to fight inflation and let the exchange rate adjust freely with undistorted capital flows.
[Friedman \(1953\)](#); [Galí and Monacelli \(2005\)](#)
- **HA real-income transmission in closed and open economies.** Monetary policy reaches constrained households through activity-induced indirect income effects. Real income channel of exchange rates.
[Kaplan, Moll and Violante \(2018\)](#); [Auclert \(2019\)](#); [Auclert, Rognlie and Straub \(2024\)](#)
- **Intermediation frictions and FX interventions.** Study exchange rates, FX policy for closing intermediation frictions and stabilizing exchange rates under capital flow volatility.
[Gabaix and Maggiori \(2015\)](#); [Fanelli and Straub \(2021\)](#); [Itskhoki and Mukhin \(2023\)](#)
- **Energy shocks and HANK.** Monetary transmission and household incidence under energy shocks.
[Blanchard and Galí \(2010\)](#); [Auclert et al. \(2024\)](#); [Chan, Diz and Kanngiesser \(2024\)](#)
- **Optimal policy in HANK.** Monetary policy redistributes; transfers target.
[Acharya, Challe and Dogra \(2023\)](#); [Dávila and Schaab \(2023\)](#); [McKay and Wolf \(2026\)](#)

A Minimal Analytical Framework

- Two-period, two-agent framework. Real model, no nominal rigidity or monetary policy. **Goal: isolate the role of FX policy under heterogeneity.**
- Open economy: small in asset and import markets; large in export market.
- **Domestic output** Y_t : produced using labor and an imported input M_t . Output is tradeable; consumed at home or exported abroad:

$$C_t + X_t = Y_t$$

- **Foreign good (dollars/consumption basket)**: not produced or consumed at home; world numeraire.
- **Imported input**: price P_t^{T*} in foreign units.
- **Asset markets**: economy accesses international asset markets, assets **denominated in foreign good units**; world real interest rate R^* .
- Economy consumes domestic good only. P_t : nominal price and price level; e_t : nominal exchange rate. **Real exchange rate**: $\epsilon_t = e_t/P_t$, relative price of foreign good in domestic units.
- **Terms of trade**: $\epsilon_t P_t^{T*}$, relative price of the imported input in domestic units.

Firms and Production Technology

- Domestic output is produced with labor and the imported input using a Leontief technology:

$$Y_t = A \cdot \min \left\{ N_t, \frac{M_t}{\theta} \right\}$$

- Cost function: $C_t = w_t N_t + \epsilon_t P_t^T * M_t$
- Cost minimization implies $M_t = \theta N_t$, $Y_t = A N_t$. Marginal cost: $\mathcal{MC}_t = w_t + \epsilon_t P_t^T * \theta$
- Zero-profit condition: $Y_t = w_t N_t + \epsilon_t P_t^T * \theta N_t$ which gives:

$$w_t = A - \epsilon_t P_t^T * \theta$$

- A unit of labor with θ units of the imported input produces A units of output, of which w_t accrues to the worker and $\epsilon_t P_t^T * \theta$ paid to buy the imported input.
- A higher import price or a real depreciation lowers real wage one-for-one to restore zero profits in equilibrium.

Export Demand and the Marshall-Lerner Condition

- Downward-sloping demand for exports. World demand for the domestic good:

$$X_t = \zeta \epsilon_t^{1+\chi}$$

- $(1 + \chi)$ is the price elasticity of export demand, with $\chi > 0$: relatively price-elastic demand.
- Export revenue in foreign units:

$$\frac{X_t}{\epsilon_t} = \zeta \epsilon_t^{\chi}$$

- **Marshall-Lerner condition:** $0 < \chi < \infty$, a real depreciation required to generate higher foreign purchasing power.

Households

- Two household types: **Hand-to-mouth**: mass $\lambda \in [0, 1)$. **Ricardian**: mass $1 - \lambda$.
- Consumption: C_t^P , C_t^R . Ricardians participate in asset markets. Asset position: f^* .
- Both types supply N units of labor inelastically.
- FX policy**: capital control tax τ on Ricardian positions, rebated lump-sum T .
- No fiscal redistribution: HtMs receive no direct transfers. Isolates how FX policy reaches constrained households *indirectly*.

Ricardians (mass $1 - \lambda$)

$$\max_{C_t^R, f^*} u(C_1^R) + \beta u(C_2^R) \quad \text{s.t.}$$

$$C_1^R + (1 + \tau) \epsilon_1 f^* = w_1 N + T$$

$$C_2^R = w_2 N + \epsilon_2 R^* f^*$$

Euler equation:

$$\underbrace{\frac{u'(C_1^R)}{\beta u'(C_2^R)}}_{\text{Domestic real rate}} = \underbrace{\frac{1}{(1 + \tau)}}_{\text{Wedge}} \underbrace{R^* \frac{\epsilon_2}{\epsilon_1}}_{\text{Effective world real rate}}$$

Hand-to-Mouth (mass λ)

- No asset market access.
- Consume current incomes each period:

$$C_t^P = w_t N$$

Balance of Payments

- Period-by-period external budget constraints:

$$P_1^T \theta N + F^* = \zeta \epsilon_1^X$$

$$P_2^T \theta N = \zeta \epsilon_2^X + R^* F^* \quad F^* = \frac{f^*}{1 - \lambda}$$

- BoP identity: Trade balance = Net capital outflows $\equiv NX_t$:

$$\underbrace{NX_t}_{\text{Trade balance}} = \underbrace{\zeta \epsilon_t^X}_{\text{Export revenue}} - \underbrace{P_t^T \theta N}_{\text{Import bill}}$$

- Aggregate intertemporal budget constraint:

$$\underbrace{P_1^T \theta N + \frac{1}{R^*} P_2^T \theta N}_{\text{PDV of import bills}} = \underbrace{\zeta \epsilon_1^X + \frac{1}{R^*} \zeta \epsilon_2^X}_{\text{PDV of export revenues}}$$

- Leontief + inelastic labor supply = perfectly inelastic import demand.

A Utilitarian Ramsey Planner's Problem

- Maximize population-weighted social welfare:

$$\max_{\{\epsilon_t\}} \lambda [u(C_1^P) + \beta u(C_2^P)] + (1 - \lambda) [u(C_1^r) + \beta u(C_2^r)] \quad \text{s.t.}$$

$$NX_1(\epsilon_1) + \frac{1}{R^*} NX_2(\epsilon_2) = 0$$

where

$$Y_t^N \equiv (A - \epsilon_t P_t^{T*} \theta) N \quad \text{real labor income}$$

$$C_t^P = Y_t^N$$

$$C_t^r = Y_t^N - \frac{\epsilon_t NX_t}{1 - \lambda}$$

- Primal formulation: Ricardian Euler equation not an implementability constraint.
- Lemma.** The solution to this problem is implementable as a decentralized equilibrium with an appropriate FX policy (τ, T) .

Optimal FX Policy

- Assume $\beta R^* = 1$. Laissez-faire equilibrium: $\tau = 0$.

Proposition: Representative agent ($\lambda = 0$)

For **any** $(P_1^{T*}, P_2^{T*}) > 0$, the optimal allocation involves **no intertemporal distortions**: $\tau = 0$.

- Optimal policy is laissez-faire under RA: all agents smooth consumption perfectly given any price path.
- Special case: no ToT manipulation to internalize export market power.

Proposition: Heterogeneous agents ($\lambda \in (0, 1)$)

If $P_1^{T*} = P_2^{T*}$, then $\tau = 0$ for any λ .

If $P_1^{T*} > P_2^{T*}$, with $(\epsilon_t^{lf}, Y_t^{N,lf})$ laissez-faire and $(\epsilon_t^O, Y_t^{N,O}, \tau^O)$ the constrained efficient allocation:

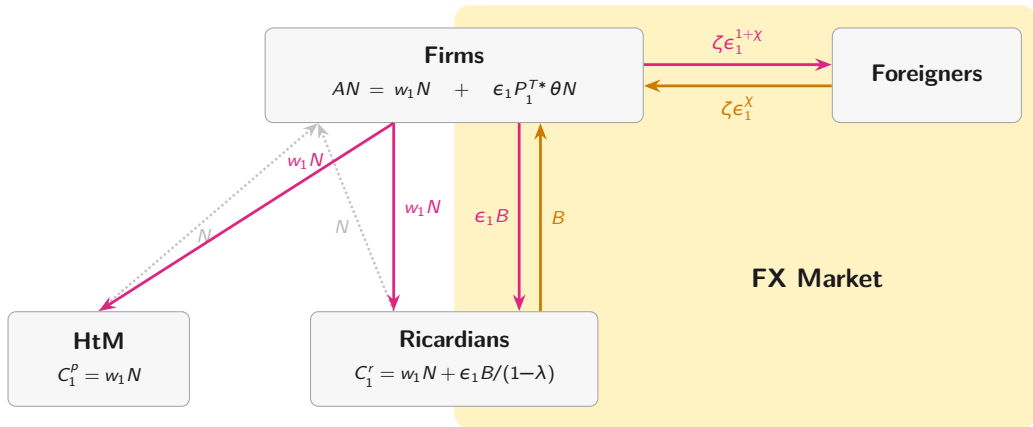
- Weaker RER co-movement:** appreciation in response to the shock: $\epsilon_1^O < \epsilon_1^{lf}$, $\epsilon_2^O > \epsilon_2^{lf}$
- Real income and ToT stabilization:** $|Y_2^{N,O} - Y_1^{N,O}| < |Y_2^{N,lf} - Y_1^{N,lf}|$, $|\epsilon_1^O P_1^{T*} - \epsilon_2^O P_2^{T*}| < |\epsilon_1^{lf} P_1^{T*} - \epsilon_2^{lf} P_2^{T*}|$
- More volatile capital flows:** $NX_1^O < NX_1^{lf} < 0 < NX_2^{lf} < NX_2^O$
- Strictly positive intertemporal wedge:** $\tau^O > 0$

What Happened?

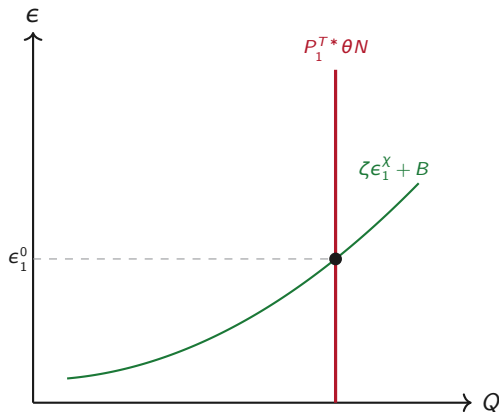
- Subsidize Ricardians to induce stronger net capital inflows in the high-price period. Import demand perfectly inelastic: capital inflows substitute for export revenues and appreciate the real exchange rate.
- Fewer exports and more resources retained at home at $t = 1$.
- At $t = 2$, after the spike has passed, must tolerate more depreciation and exports, larger capital outflows.
- **Why is this optimal? Smoother consumption path for HtM households through the shock.**
- No direct fiscal transfers: any additional borrowing induced by policy accrues to Ricardians and directly distorts their consumption path. **How does this policy deliver smoother consumption to HtM households then?**
- **The real income stabilization channel:** Capital-flow-induced RER appreciation raises real wages for all workers and **indirectly** boosts their purchasing power. Period 2 reverses this direction.
- **Marshall-Lerner condition** is the key: without this, policy cannot appreciate and boost real incomes.

Flow of Goods: The Foreign Exchange Market

- Suppose $P_1^{T*} > P_2^{T*}$ so that Ricardians are net borrowers at $t = 1$ with $B \equiv -F^*$.
- Ricardians and foreigners are net suppliers of the **foreign good** and demand the **domestic good**; firms are net demanders of the foreign good and supply the domestic good in exchange.



The FX Market

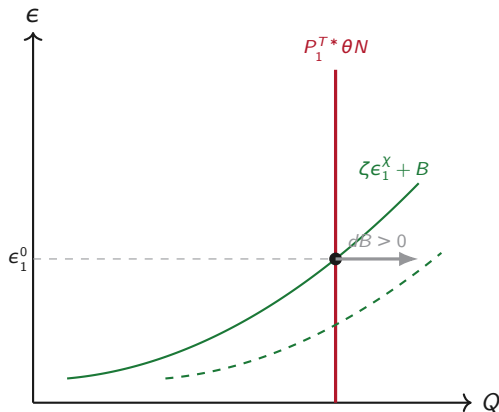


- BoP condition as foreign exchange market equilibrium: demand for and supply of the **foreign good**, with the real exchange rate as the market-clearing price:

$$\underbrace{P_1^T * \theta N}_{\text{Demand}} = \underbrace{\zeta \epsilon_1^X + B}_{\text{Supply}}$$

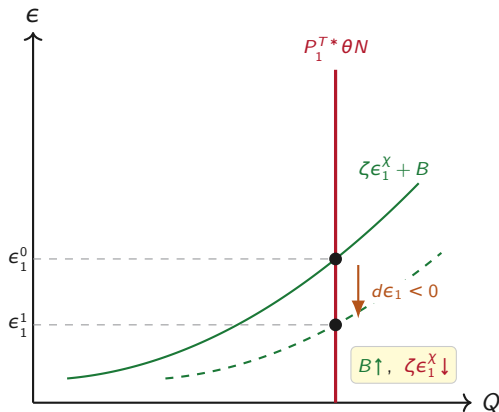
- Import demand perfectly inelastic.
- Marshall-Lerner condition:** export revenue supply is upward-sloping: more foreign good earned through a real depreciation.

The FX Market: FX Policy Shifts Capital Inflows



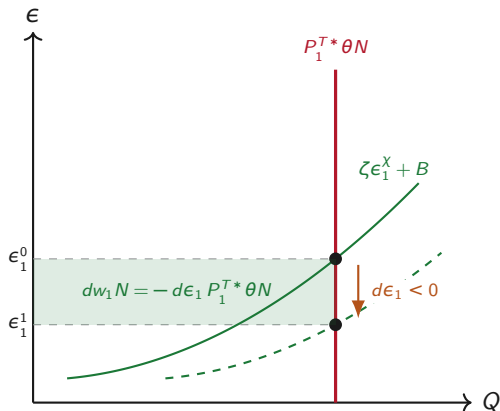
- **FX policy** induces stronger net capital inflows: Ricardians borrow more, $dB > 0$.
- The supply of foreign good in the FX market increases.
- More foreign purchasing power available to pay the import bill.

The FX Market: Real Appreciation



- Excess supply makes Ricardians bid down the real exchange rate.
- Foreigners **crowded out**: export revenues fall.
- The RER appreciates: $d\epsilon_1 < 0$.

The FX Market: Real Wage Gains



- A real appreciation lowers marginal costs for firms: positive profit opportunities arise that induce them to expand their import demand.
- Labor and imports are perfect complements: labor demand expands as firms compete for workers. With labor supply given, firms **bid the real wage up** until zero profits are restored: $w_1 = A - \epsilon_1 P_1^T * \theta$.
- **Every worker gains** through an indirect real income effect: capital inflows appreciate the RER and boost the real income of every worker, including the constrained.
- Marshall-Lerner elasticity crucial: income effect disappears as $\chi \rightarrow \infty$. Stronger as $\chi \rightarrow 0$.

Taking Stock: FX Policy and Real Income Stabilization

- Borrowing constraints and the Marshall-Lerner condition turn the exchange rate from a **shock absorber into an inflation amplifier**: given an essential-import price spike, constrained households face amplified current real income losses due to a concurrent depreciation of the real exchange rate.
- **The real-income stabilization channel of exchange rate policy**: induce stronger capital inflows than laissez-faire to lean against a concurrent depreciation during the spike. Real wages rise for all workers: gains reach constrained households through indirect income effects.
- Policy re-times, does not eliminate the PDV loss: tolerates more depreciation, higher exports and lower real wages after the spike. Achieves a smoother consumption path for constrained households.
- Works by driving **interest parity wedges**. Optimal real rate endogenous.
- Does not require nominal rigidities and works independently of the level of activity: output and employment.

Adding Elastic Labor Supply and Monetary Policy

- Introduce elastic labor supply: N_t hours of employment. Both household types work the same amount of time, with disutility $v(N_t)$.
- Households do not choose hours. Nominal aggregate demand and monetary policy together determine aggregate labor supply.
- **Rigid nominal wage** $\overline{W}$. Phillips curve: $P_t = \overline{W}/w_t$.
- **Monetary policy** sets the money supply, $\mathcal{M}_t$, and the exchange equation determines nominal aggregate demand:

$$\mathcal{M}_t = P_t Y_t = P_t A N_t$$

- Monetary policy and the Phillips curve together determine the price level and employment. Real wage determined by firms' ZPC as before.
- With rigid nominal wages: a lower real wage is equivalent to a higher price level and vice versa.

The Ramsey Planner's Problem Revisited

- The utilitarian Ramsey planner now sets monetary and FX policy jointly:

$$\max_{\{N_t, \epsilon_t\}} \lambda [u(C_1^p) + \beta u(C_2^p)] + (1 - \lambda) [u(C_1^r) + \beta u(C_2^r)] - [v(N_1) + \beta v(N_2)] \quad \text{s.t.}$$
$$NX_1(\epsilon_1, N_1) + \frac{1}{R^*} NX_2(\epsilon_2, N_2) = 0$$

where

$$Y_t^N = (A - \epsilon_t P_t^{T*} \theta) N_t, \quad C_t^p = Y_t^N, \quad C_t^r = Y_t^N - \frac{\epsilon_t NX_t}{1 - \lambda}$$

- Primal formulation: Ricardian Euler equation and the nominal exchange equations are not implementability constraints.
- Nominal rigidity introduces non-neutrality: has no real costs by itself, but allows monetary policy to distort the **labor wedge** and influence the real economy through employment and output.
- **Lemma.** The solution to this problem is implementable as a decentralized equilibrium with an appropriate fiscal-monetary policy $(\tau, T, \mathcal{M}_t)$.

Joint Optimal Policy: RA

Proposition: Representative agent ($\lambda = 0$)

For any $(P_1^{T*}, P_2^{T*}) > 0$, the optimal allocation involves no intertemporal distortions: $\tau = 0$. The labor wedge is distorted to account for the **static terms-of-trade correction**:

$$v'(N_t) = u'(C_t) \left(w_t - \frac{\epsilon_t P_t^{T*} \theta}{\chi} \right)$$

- Leontief: ToT externality corrected through labor wedge; planner internalizes static market power over exports: each additional unit of labor requires imported inputs, which depreciates the RER and worsens the terms of trade.
- Laissez-faire FX policy: all households smooth consumption optimally, no reason for policy to distort financial flows.

Joint Optimal Policy: HA

Proposition: Heterogeneous agents ($\lambda \in (0, 1)$)

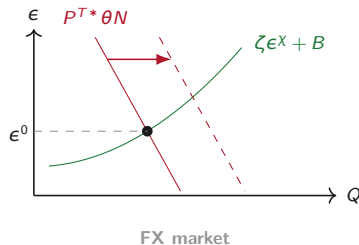
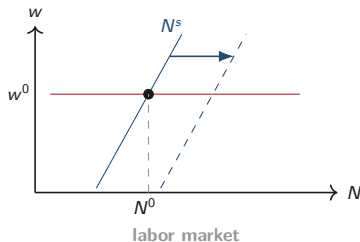
- $P_1^{T*} = P_2^{T*}$: no intertemporal distortions; labor wedge corrects for the terms-of-trade effect.
- $P_1^{T*} > P_2^{T*}$: positive intertemporal wedge, $\tau > 0$; labor wedge corrects for both the static terms-of-trade effect and a **redistributive price effect**:

$$v'(N_t) = \bar{u}'_t \left(w_t - \frac{\epsilon_t P_t^{T*} \theta}{\chi} \right) + \lambda [u'(C_t^p) - u'(C_t^r)] N X_t \frac{d\epsilon_t}{dN_t} \Big|_{N X_t}$$

- When $P_1^{T*} > P_2^{T*}$, the redistributive price effect creates an additional motive to restrain employment and imports: optimal monetary policy more contractionary than an RA-equivalent rule.
- FX policy implements a positive intertemporal wedge as before: subsidizes Ricardians to induce stronger capital inflows during the high-price period.
- With FX policy active, monetary policy's role purely static: does not influence capital flows.

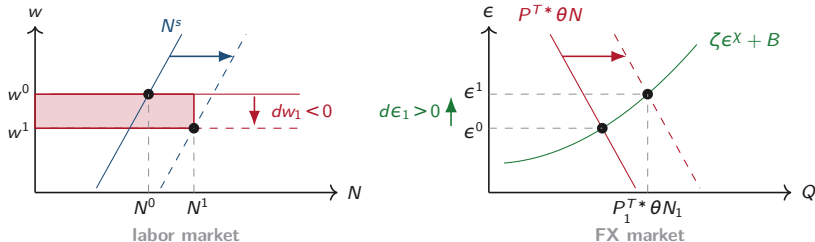
A Marginal Monetary Expansion: More Employment, More Imports

- Suppose $P_1^{T*} > P_2^{T*}$, Ricardians borrow: $B = -NX_1 > 0$. Consider a monetary expansion: $dN_1 > 0$.
- Labor supply expands: downward pressure on real wages lowers marginal costs and opens profit opportunities. Firms expand to hire the additional labor.
- Import demand expands.** Labor and imports are perfect complements: firms also demand more imported inputs.
- Hold $B = -NX_1$ fixed: additional imports financed by current exports. General equilibrium with FX policy response endogenous: at the margin, indifferent between financing additional imports by current exports or capital inflows. Envelope theorem argument: employment's marginal impact purely static.



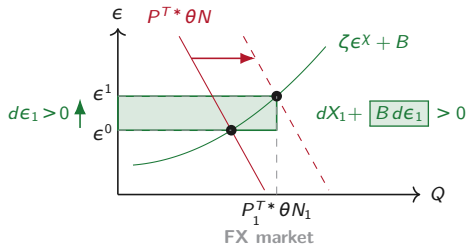
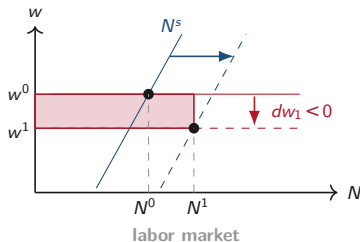
A Marginal Monetary Expansion: Depreciation and the Terms of Trade

- Marshall-Lerner: foreigners only supply the requisite purchasing power at a lower world price. RER **depreciates**, $d\epsilon_1 > 0$ and exports rise, $dX_1 > 0$.
- Depreciation raises the marginal cost of imports. Zero profits: as the RER depreciates, real wage falls for every worker, $dw_1 < 0$.
- **Terms-of-trade externality**: employment expansion increases exports by depreciating the RER, and lowers the real wage. Firms do not internalize this, planner does; present even in an RA economy.



A Marginal Monetary Expansion: Redistributive Price Effect

- An import-induced depreciation also creates a **pecuniary** benefit for Ricardians: allows them to buy more of the domestic good, $B d\epsilon_1$, for the given amount of net foreign unit inflows, B .
- Asymmetric gains:** real wages lost by workers finance this gain. In RA, this effect cancels out. In HA, **all borrowers are workers but not all workers are borrowers**.
- HtMs bear a lower wage: a net endogenous **transfer from HtMs to Ricardians**.
- Planner internalizes this asymmetric **redistributive price effect** of a monetary expansion and has incentive to restrain expansion beyond an RA-equivalent rule in the high import-price period.



Taking Stock: Joint Optimal Policy

- Relative to RA: during spike episodes, **FX policy induces stronger capital inflows** and **monetary policy is more contractionary**.
- FX policy distorts the **intertemporal wedge**. Monetary policy distorts the static **labor wedge** within each period and under employment-import complementarity, influences import demand.
- Both influence the FX market through different channels and thereby influence the equilibrium RER.
- FX policy seeks appreciation and higher real wages through capital inflows. Realized appreciation depends on how much monetary policy allows employment to expand.
- Feasible: FX policy induces capital inflows, monetary policy allows higher employment and imports, but must tolerate more depreciation and lower real wages; ToT externality hurts workers, redistributive price effects hurt constrained households.
- Also feasible: monetary policy restrains employment. FX-policy-induced capital inflows appreciate the RER and raise real wages: lower prices under rigid nominal wages.
- **Optimal**: real income gains to constrained through **higher wages, restrained employment**.

Monetary Policy in Isolation

- Without FX policy, monetary policy faces the Ricardian Euler equation, i.e., the **interest parity condition** as an implementability constraint:

$$\frac{u'(C_1^r)}{\beta u'(C_2^r)} = R^* \frac{\epsilon_2}{\epsilon_1}$$

- Ricardian budget constraints and the Euler equation imply the Ricardian **saving function**:

$$F^* = F^*(\epsilon_1, \epsilon_2, N_1, N_2)$$

- Monetary policy sets the employment path, $\{N_1, N_2\}$; capital flows and the RER path follow as a consequence of balance of payments:

$$\begin{aligned} P_1^T \theta N_1 + F^*(\epsilon_1, \epsilon_2, N_1, N_2) &= \zeta \epsilon_1^X \\ P_2^T \theta N_2 &= \zeta \epsilon_2^X + R^* F^*(\epsilon_1, \epsilon_2, N_1, N_2) \end{aligned}$$

- FX policy breaks this dependency: moves F^* independently of the employment path through interest parity wedges: tighter grip over the RER, independent of the level of real activity.

Quantitative Model

- Baseline: two-agent infinite-horizon framework.
- Segmented asset markets: Ricardians cannot access international assets directly. Domestic assets: nominal units; international assets: foreign-good units.

- Ricardians solve:

$$\max_{\{C_t^r, B_{t+1}^{M,r}\}} \sum_{t=1}^{\infty} \beta^{t-1} [u(C_t^r) - v(N_t)] \quad \text{s.t.}$$

$$P_t C_t^r + B_{t+1}^{M,r} = W_t N_t + R_t^M B_t^{M,r} + T_t$$

- Optimality:

$$\frac{u'(C_t^r)}{\beta u'(C_{t+1}^r)} = R_{t+1}, \quad \lim_{T \rightarrow \infty} \frac{B_{T+1}^r}{\prod_{s=2}^T R_s} = 0, \quad B_{t+1}^r = \frac{B_{t+1}^{M,r}}{P_t}$$

- $R_{t+1} = \frac{R_{t+1}^M}{\Pi_{t+1}}$: real interest rate in domestic units; $\Pi_{t+1} = \frac{P_{t+1}}{P_t}$ gross inflation rate.

Nominal Wage Rigidity and the Labor Wedge

- Adapted from Erceg, Henderson and Levin (2000) and Auclert, Rognlie and Straub (2024).
- Final labor service produced by an aggregator using $k \in [0, 1]$ tasks.
- For each task k , a union sets the nominal wage $W_t(k)$ for a continuum of workers.
- Each union faces a **private utility cost** of changing the nominal wage.
- Wage Phillips curve with symmetric unions:

$$v'(N_t) = (1 - \tau_t^N) w_t \bar{u}'_t,$$
$$1 - \tau_t^N = \frac{1}{\mu_w} + \frac{\left(\frac{w_t}{w_{t-1}} \Pi_t - 1 \right) \frac{w_t}{w_{t-1}} \Pi_t - \beta \left(\frac{w_{t+1}}{w_t} \Pi_{t+1} - 1 \right) \frac{w_{t+1}}{w_t} \Pi_{t+1}}{\frac{\varepsilon_w}{\psi_{nr}} w_t N_t \bar{u}'_t}$$

International Financial Intermediaries

- Adapted from [Gabaix and Maggiori \(2015\)](#).
- Take positions in domestic and international asset markets s.t. a balance sheet constraint:
$$\frac{F_{t+1}}{e_t} + F_{t+1}^* = 0.$$
- Intermediary returns:

$$V_{t+1} = \left[R^* F_{t+1}^* + R_{t+1}^M \frac{F_{t+1}}{e_{t+1}} \right] = \left[R^* - R_{t+1}^M \left(\frac{e_t}{e_{t+1}} \right) \right] F_{t+1}^*$$

- Seek to maximize returns subject to a limited commitment friction:

$$\max \frac{1}{R^*} V_{t+1} \quad \text{s.t.} \quad \frac{1}{R^*} V_{t+1} \geq \Gamma F_{t+1}^{*2}$$

- Optimal asset position a function of the interest parity gap:

$$F_{t+1}^* = \frac{1}{\Gamma} \left[1 - \frac{R_{t+1}^M}{R^*} \left(\frac{e_t}{e_{t+1}} \right) \right] = \frac{1}{\Gamma} \left[1 - \left(\frac{R_{t+1}}{R^*} \frac{e_t}{e_{t+1}} \right) \right]$$

Fiscal-Monetary Policy and Accounting

- Monetary policy sets the money growth rate. Nominal aggregate demand given by the exchange equation:

$$g_t^{\mathcal{M}} = \Pi_t \frac{Y_t}{Y_{t-1}}$$

Equivalently, monetary policy determines the ex post real rate R_1 and forward-looking nominal interest rates:

$$R_{t+1}^M = R_{t+1} \Pi_{t+1} \quad \text{for } t \geq 1$$

- Fiscal policy: domestic and foreign asset positions, with budget constraint:

$$B_{t+1}^{M,G} + e_t B_{t+1}^{G*} + (1-\lambda) T_t = R_t^M B_t^{M,G} + e_t R^* B_t^{G*}$$

$$\lim_{T \rightarrow \infty} \frac{B_{T+1}^G}{\prod_{s=2}^T R_s} = 0, \quad \lim_{T \rightarrow \infty} \frac{B_{T+1}^{G*}}{R^{*T-1}} = 0, \quad B_{t+1}^G \equiv \frac{B_{t+1}^{M,G}}{P_t}$$

- Bond-market clearing:

$$B_{t+1} + B_{t+1}^G = \epsilon_t F_{t+1}^*$$

- Balance of payments:

$$P_t^{T*} M_t + F_{t+1}^* + B_{t+1}^{G*} = \zeta \epsilon_t^X + R^* (F_t^* + B_t^{G*}) - R^* \Gamma F_t^{*2}$$

Ramsey Problem

$$\max_{\{N_t, \epsilon_t, \tilde{R}_{t+1}, C_t^r, C_t^p\}_{t=1}^{\infty}} \sum_{t=1}^{\infty} \beta^{t-1} [\lambda u(C_t^p) + (1-\lambda)u(C_t^r) - v(N_t)] \quad \text{s.t.}$$

$$\underbrace{\sum_{t=1}^{\infty} \frac{\frac{1}{\Gamma} \left(1 - \frac{\tilde{R}_{t+1}}{R^*}\right)^2}{R^{*t-1}}}_{\text{carry trade cost}} + \sum_{t=1}^{\infty} \frac{P_t^{T*} \theta N_t}{R^{*t-1}} = \sum_{t=1}^{\infty} \frac{\zeta \epsilon_t^{\chi}}{R^{*t-1}} + R^* B_1^{G*} + R_1 \frac{\epsilon_0 F_1^*}{\epsilon_1}$$

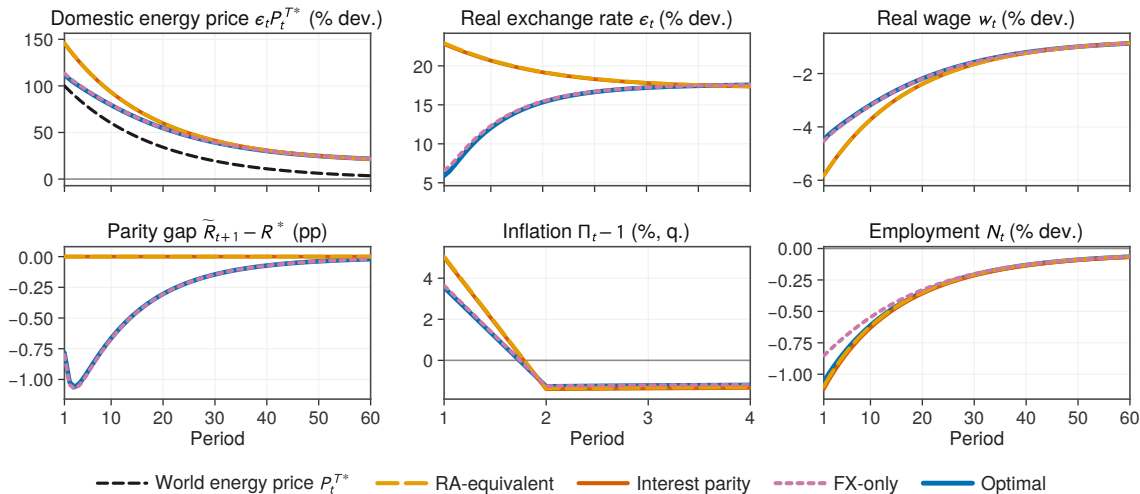
$$\tilde{R}_{t+1} = \frac{u'(C_t^r) \epsilon_t}{\beta u'(C_{t+1}^r) \epsilon_{t+1}}, \quad C_t^r = A_t N_t - \frac{\zeta \epsilon_t^{1+\chi}}{1-\lambda} + \frac{\lambda}{1-\lambda} \epsilon_t P_t^{T*} \theta N_t, \quad C_t^p = (A_t - \epsilon_t P_t^{T*} \theta) N_t$$

- Qualitatively the same primal problem as the analytical framework, except the **carry cost** term in IBC: interest parity wedges are costly in the form of carry trade costs from intermediary profits.
- Analytical framework nested: $\Gamma \rightarrow 0$, the same primal problem, decentralized with capital controls.
- **Lemma.** The solution to this problem is implementable as a decentralized equilibrium.
- Phillips curve not an implementability constraint: costless non-neutrality; inflation path implements the optimal labor wedge to satisfy the Phillips curve.

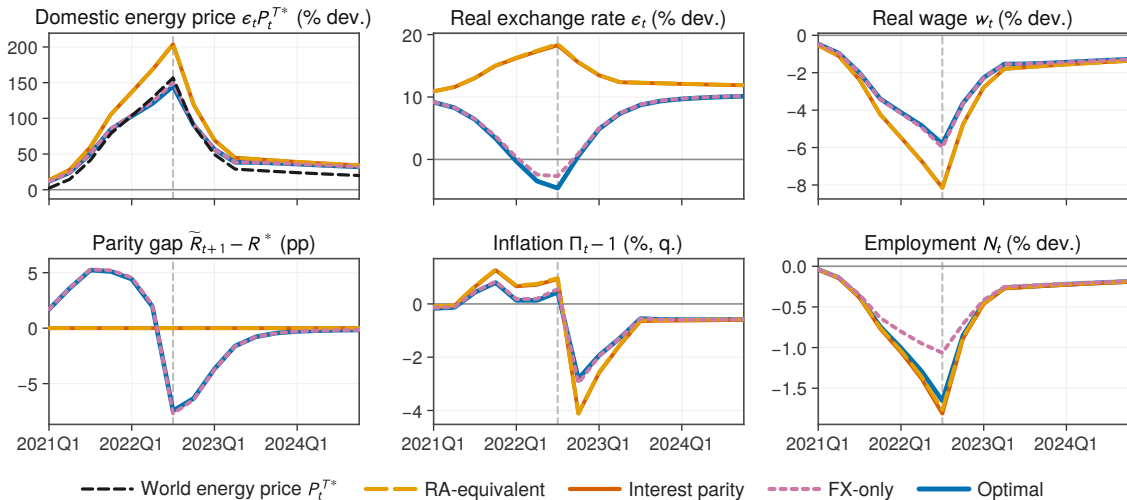
Application: Energy Price Shocks

- Essential import: fossil fuels. Calibrated to Japan: 98% import dependence; over 4% of output spent on fossil fuels in 2022.
- Compare jointly optimal policy under perfect foresight to three counterfactual policies:
 - **RA-equivalent**: no intertemporal distortions; labor wedge corrects terms-of-trade distortion.
 - **Interest parity**: FX policy closes the financial intermediation friction; implements optimal monetary policy in isolation.
 - **FX-only**: intertemporal stance free; labor wedge corrects RA-like terms-of-trade distortion only.
- Policy response under two import price paths:
 - **MIT shock**: perfect-foresight impulse response.
 - **2022 spike**: response to a deterministic 2022-like energy-price path.

Transitory Spike



2022 Spike



2022 Spike: Policy Comparison

2021–23 path	RA-equiv.	Interest parity	FX-only	Optimal
Peak energy-price amplification factor	1.30	1.30	0.96	0.92
Peak real-wage loss undone (%)	–	0.27	26.64	29.03
Peak domestic energy price (% lower)	–	0.18	17.88	19.49
Avg. quarterly inflation through peak (%)	0.60	0.58	0.28	0.23

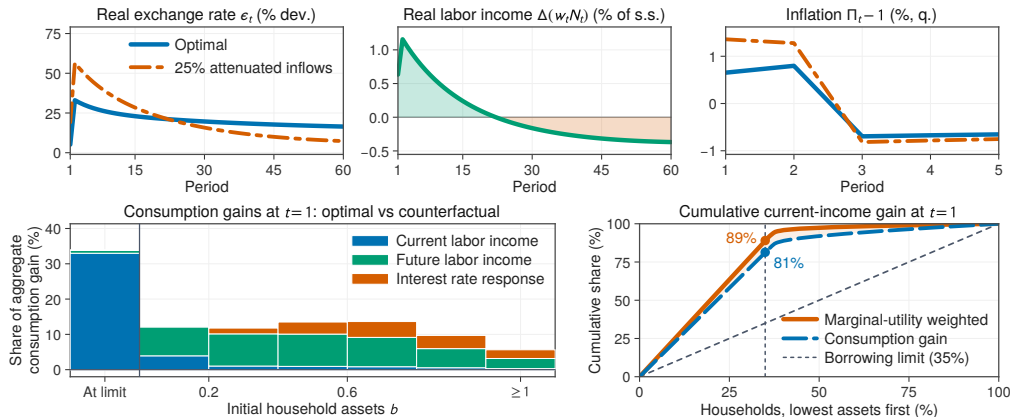
- At the peak of the crisis, heterogeneity accounts for 71.06% of the contemporaneous depreciation under RA-equivalent policy relative to a world with no HtM households.
- Monetary policy set optimally, with the intertemporal stance suppressed, does little on its own in protecting HtM households: faces the interest parity condition.
- **FX policy leads with real income protection and disinflation. Monetary policy supports with a disinflationary stance.**
- Optimal policy re-times when the country pays for real income losses, does not eliminate them: present value of aggregate consumption differs by only 0.0005% between optimal and RA-equivalent policies.

Heterogeneous Agents with Idiosyncratic Risk

- Beyond the two-agent model: Bewley-Huggett-Aiyagari framework with a continuum of households facing idiosyncratic risk, incomplete markets and ad hoc borrowing constraints.
- No Fisherian revaluation incentives: inflation-indexed domestic asset markets. No lump-sum transfers: FX policy affects prices by taking positions in domestic and foreign asset markets only.
- **A distributional time-inconsistency problem:** Planner has incentive to lower the real rate and increase inflation to redistribute towards high-MPC households ([Dávila and Schaab 2023](#)). Both FX policy and monetary policy prone to a time-inconsistency problem.
- Solve jointly optimal nonlinear perfect-foresight policies under **timeless commitment**: planner bound by past promises associated with household Euler equations encoded in Lagrange multipliers acting as promise-keeping state variables ([Marcet and Marimon 2019](#)). Computational tractability: promise-keeping state policy function is affine in idiosyncratic state variables.
- Policy exercise: analyze FX policy's indirect income effects by comparing optimal policy with a counterfactual rule that implements a 25% attenuated net capital flow in the shock impact period.

Heterogeneous Agents: The Income Effect across the Distribution

- Optimal FX policy vs counterfactual: optimal policy appreciates and tilts real incomes towards high-import-price periods. Consumption response differs across the distribution.
- Current real-income gains are valued most near the borrowing limit.



Conclusion

- Borrowing constraints and the Marshall-Lerner condition turn the RER from a **shock absorber into an inflation amplifier**: amplifies current real-income losses for constrained households during episodes of essential-import price spikes.
- A laissez-faire approach to exchange rate management is costly for constrained households. Optimal policy should **lean against a contemporaneous real depreciation** during these episodes.
- **The real-income stabilization channel**: capital inflows appreciate the RER and raise real wages for all workers. Gains reach constrained households through **indirect income effects**.
- With FX policy active, the role for monetary policy changes and becomes supportive. As opposed to a typical HANK prescription, optimal monetary policy is more contractionary than an RA-equivalent rule: a redistributive price effect, attributable to the Marshall-Lerner condition, is at play.
- **FX policy becomes an anti-inflation policy**. In a monetary economy, real income protection arrives in the form of lower inflation. FX policy is more effective than monetary policy in addressing import-price-spike-induced inflation.
- Monetary policy in isolation is limited by the interest parity restriction: it cannot drive intertemporal wedges in a small open economy. Most real income protection power of policy lives here.

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Appendix

Equilibrium Definition

Given an exogenous sequence $\{P_t^{T*}\}$ and exchange rate policy (τ, T) , an equilibrium consists of $\{C_t^P, C_t^r, F^*, w_t, \epsilon_t\}$ such that:

- HtM households consume current income: $C_t^P = w_t N$
- Ricardian households optimize: $u'(C_1^r)\epsilon_1(1 + \tau) = \beta R^* u'(C_2^r)\epsilon_2$
- Firms optimize: $w_t = A - \epsilon_t P_t^{T*} \theta$
- Fiscal budget holds: $T = \tau \epsilon_1 \frac{F^*}{1 - \lambda}$
- Domestic goods market clears:

$$\lambda C_t^P + (1 - \lambda) C_t^r + \zeta \epsilon_t^{1+\chi} = AN$$

- Balance of payments:

$$P_1^{T*} \theta N + F^* = \zeta \epsilon_1^\chi, \quad P_2^{T*} \theta N = \zeta \epsilon_2^\chi + R^* F^*$$

Optimality Conditions

Optimal intertemporal wedge:

$$\epsilon_1 u'(C_1^r) + \Omega_1 = \beta R^* [\epsilon_2 u'(C_2^r) + \Omega_2]$$

$$\Omega_t \equiv \frac{\lambda [u'(C_t^p) - u'(C_t^r)] \left(\frac{\epsilon_t}{\chi} - NX_t \frac{d\epsilon_t}{dNX_t} \right)}{\frac{1+\chi}{\chi}}$$

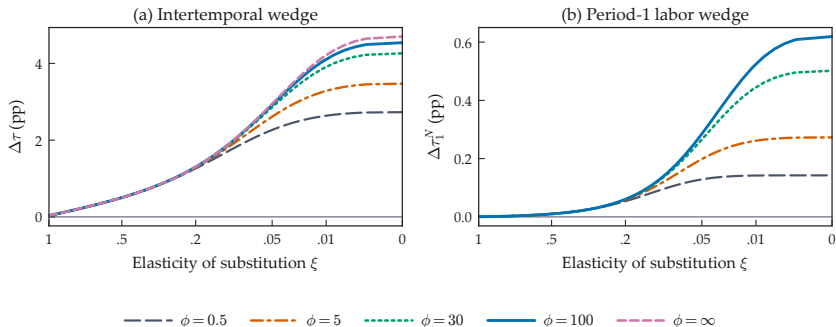
$$\tau = \frac{\Omega_1 - \beta R^* \Omega_2}{\epsilon_1 u'(C_1^r)}$$

Optimal labor wedge:

$$v'(N_t) = \bar{u}'_t \left(w_t - \frac{\epsilon_t P_t^{T*} \theta}{\chi} \right) + \lambda [u'(C_t^p) - u'(C_t^r)] NX_t \frac{d\epsilon_t}{dN_t} \Big|_{NX_t}$$

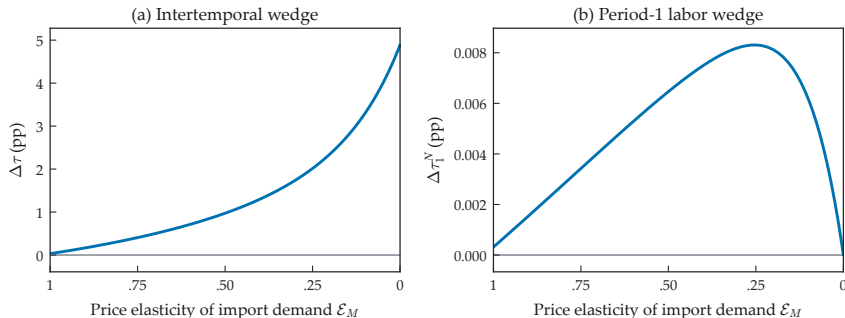
$$\tau_t^N = \frac{1}{\chi} \frac{\epsilon_t P_t^{T*} \theta}{w_t} - \frac{\lambda [u'(C_t^p) - u'(C_t^r)] NX_t \frac{d\epsilon_t}{dN_t} \Big|_{NX_t}}{\bar{u}'_t w_t}$$

Comparative Statics: CES Production



- Generalize production to CES:
$$Y_t = A_t \left[(1 - \alpha) N_t^{\frac{\xi-1}{\xi}} + \alpha \left(\frac{M_t}{\theta} \right)^{\frac{\xi-1}{\xi}} \right]^{\frac{\xi}{\xi-1}}.$$
- Comparative statics with respect to the price elasticity of import demand. Plotted case: $P_1^{T*} > P_2^{T*}$, optimal wedge differences between $\lambda > 0$ and $\lambda = 0$ economies. Increasing as $\xi \rightarrow 0$ and $\phi \rightarrow \infty$: **essential imports strengthen both policy motives** under heterogeneity.

Comparative Statics: Imports as Final Goods



- Imported good as a final good with Stone-Geary preferences: $u(C, M) = \frac{[C^{1-\omega}(M-\underline{M})^\omega]^{1-\sigma}}{1-\sigma}$, $M \geq \underline{M}$.
- Comparative statics with respect to the price elasticity of import demand. Plotted case: $P_1^{T*} > P_2^{T*}$. Optimal wedge differences between $\lambda > 0$ and $\lambda = 0$ economies.
- **FX policy's role increasing with essentiality:** $\Delta\tau$ increases as import demand becomes inelastic.
- $\Delta\tau_1^N$ small, weaker role for monetary policy. Perfectly inelastic demand: heterogeneity adds no role.

- The competitive labor packer combines differentiated tasks into final labor:

$$N_t = \left(\int_0^1 N_{k,t}^{\frac{\varepsilon_w - 1}{\varepsilon_w}} dk \right)^{\frac{\varepsilon_w}{\varepsilon_w - 1}}, \quad \varepsilon_w > 1$$

- It solves

$$\min_{\{N_{k,t}\}_{k \in [0,1]}} \int_0^1 W_{k,t} N_{k,t} dk \quad \text{s.t.} \quad \left(\int_0^1 N_{k,t}^{\frac{\varepsilon_w - 1}{\varepsilon_w}} dk \right)^{\frac{\varepsilon_w}{\varepsilon_w - 1}} \geq N_t$$

- Cost minimization yields

$$N_{k,t} = \left(\frac{W_{k,t}}{W_t} \right)^{-\varepsilon_w} N_t, \quad W_t = \left(\int_0^1 W_{k,t}^{1-\varepsilon_w} dk \right)^{\frac{1}{1-\varepsilon_w}}$$

Unions' Wage-Setting Problem and the WPC

- Taking the task-demand schedule as given, union k chooses its nominal wage sequence to solve

$$\max_{\{W_{k,t+s}\}_{s \geq 0}} \sum_{s \geq 0} \beta^s \left[\lambda u(C_{t+s}^p) + (1-\lambda)u(C_{t+s}^r) - v(N_{t+s}) - \frac{\psi_{nr}}{2} \left(\frac{W_{k,t+s}}{W_{k,t+s-1}} - 1 \right)^2 \right]$$

subject to

$$N_{k,t+s} = \left(\frac{W_{k,t+s}}{W_{t+s}} \right)^{-\varepsilon_w} N_{t+s}.$$

- Each union is infinitesimal and takes W_{t+s} and N_{t+s} as given.
- In a symmetric equilibrium, $W_{k,t} = W_t$, $N_{k,t} = N_t$ for every k . Wage Phillips curve:

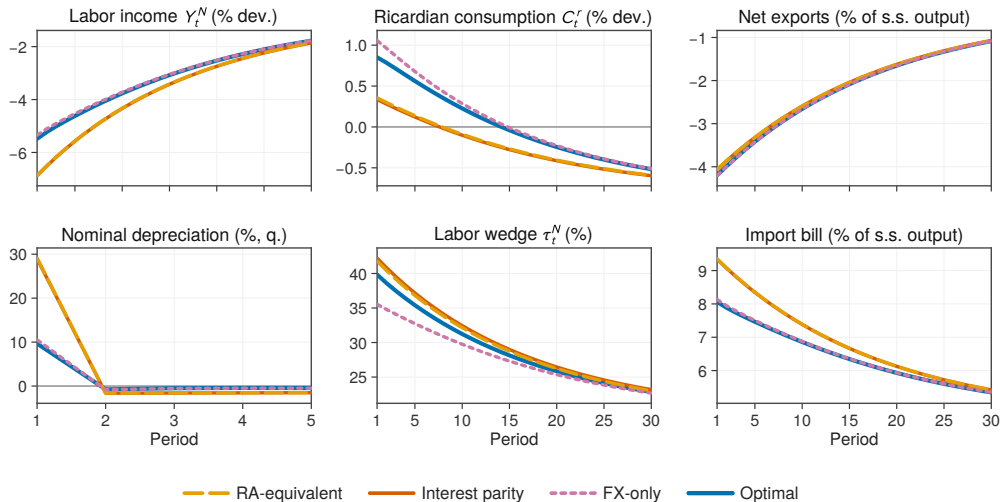
$$v'(N_t) = (1 - \tau_t^N) w_t \bar{u}'_t,$$

$$1 - \tau_t^N = \frac{1}{\mu_w} + \frac{\left(\frac{w_t}{w_{t-1}} \Pi_t - 1 \right) \frac{w_t}{w_{t-1}} \Pi_t - \beta \left(\frac{w_{t+1}}{w_t} \Pi_{t+1} - 1 \right) \frac{w_{t+1}}{w_t} \Pi_{t+1}}{\frac{\varepsilon_w}{\psi_{nr}} w_t N_t \bar{u}'_t}$$

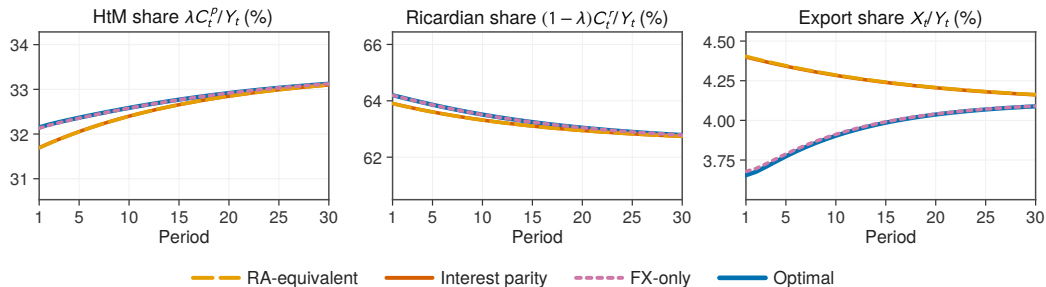
Quarterly calibration for Japan

	Parameter	Value	Target or source
R^*	Gross world interest rate	1.0021	Net exports of -0.5% of GDP
β	Discount factor	0.9979	$\beta R^* = 1$
σ	Inverse intertemporal elasticity	4.0	Intertemporal elasticity of 0.25
η	Labor-disutility scale	0.9261	Steady-state employment normalization
ϕ	Inverse Frisch curvature	29.36	Output falls 1% on impact after a 100% import-price rise in the RA economy
λ	HtM household share	0.35	Kaplan, Violante and Weidner (2014) ballpark for US
ζ	Export demand scale	1.9758	Fossil-fuel import bill of 4%
χ	Marshall-Lerner elasticity	0.25	Modest short-run response: Rose (1991) ; Bahmani-Oskooee and Ratha (2004)
ρ_{PT^*}	Import-price persistence	0.945	AR(1) on detrended IMF energy price index (International Monetary Fund 2026)
Γ	Intermediation friction	9.0	Fanelli and Straub (2021)
ε_w	Wage substitution elasticity	6.129	Core CPI inflation of 0.097% per quarter (2014Q4–2019Q4) in RA steady state
ψ_{nr}	Nominal wage rigidity	10598.1	Core CPI inflation of 0.411% per quarter (2020Q4–2022Q3) in RA 2022 spike
$B_{ss}^{G^*}$	Public foreign assets	60.0	Net foreign assets of 60% of annual GDP

Transitory Spike: Additional Outcomes

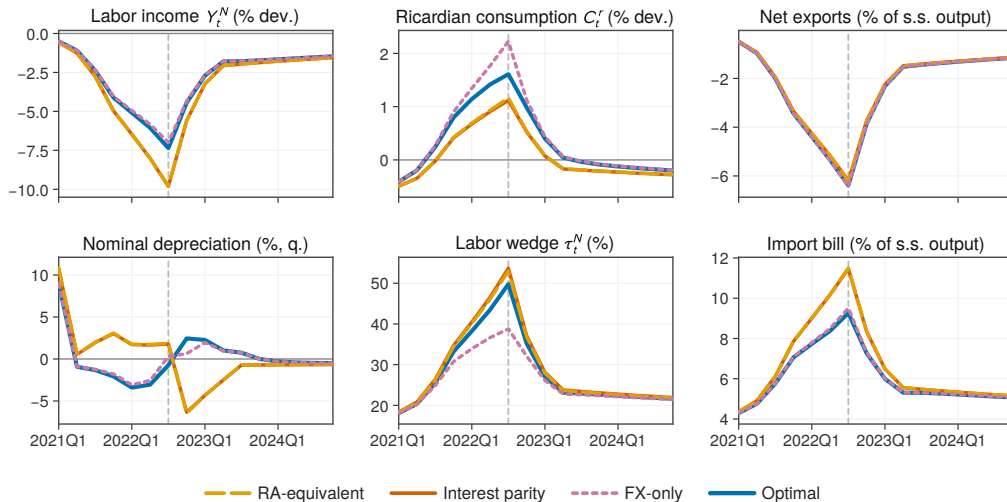


Transitory Spike: Output Shares

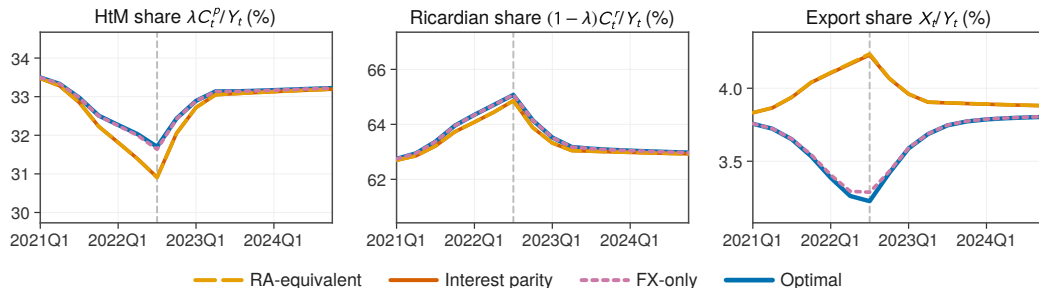


- During the spike, optimal policy retains more output for domestic consumption and reduces the export share. HtM share goes up due to indirect income effects.

2022 Spike: Additional Outcomes



2022 Spike: Output Shares



- Around the peak of the crisis, optimal policy lowers the export share and retains more output for domestic consumption. HtM share goes up due to indirect income effects.

Heterogeneous Agent Framework

- Unit mass of households, autonomous consumption requirement z_t^i .
- $z_{t+1}^i \sim Z(\cdot | z_t^i)$, invariant π_z , fixed aggregate mean $\int z d\pi_z = \bar{z}$: no aggregate risk.
- Inflation-indexed domestic asset market. Common labor supply N_t as before.
- Household problem:

$$\max_{\{c_t^i, b_{t+1}^i\}} \mathbb{E}_1 \sum_{t=1}^{\infty} \tilde{\beta}^{t-1} [u(c_t^i) - v(N_t)]$$

$$c_t^i + b_{t+1}^i = R_t b_t^i + w_t N_t - z_t^i + (R_t - 1)\underline{b}, \quad b_{t+1}^i \geq 0$$

- Optimality conditions:

$$u'(c_t^i) \geq \tilde{\beta} R_{t+1} \mathbb{E}_t u'(c_{t+1}^i), \quad b_{t+1}^i [u'(c_t^i) - \tilde{\beta} R_{t+1} \mathbb{E}_t u'(c_{t+1}^i)] = 0$$

- FX policy with no transfers: public agent subject to period budgets and no-Ponzi conditions.

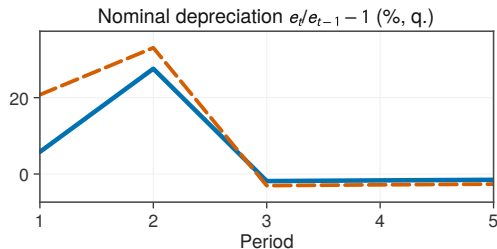
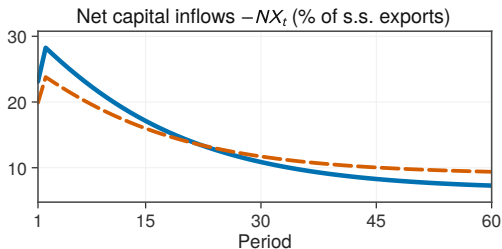
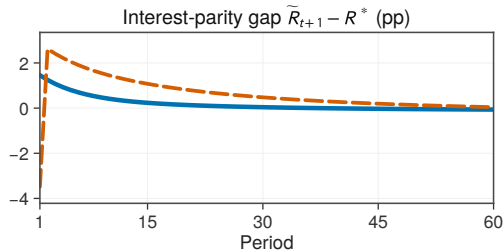
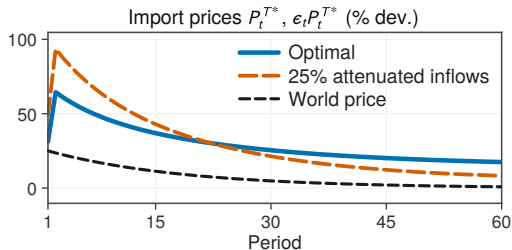
$$B_{t+1}^G + \epsilon_t B_{t+1}^{G*} = R_t B_t^G + \epsilon_t R^* B_t^{G*}$$

Heterogeneous Agents: Ramsey Problem under Timeless Commitment

$$\begin{aligned}
 & \max_{\{c_t^i, b_{t+1}^i, N_t, \epsilon_t, R_{t+1}\}_{t \geq 1}} \mathbb{E}_1 \sum_{t=1}^{\infty} \beta^{t-1} \left[\int u(c_t^i) di - v(N_t) \right] - \frac{\tilde{\beta}}{\beta} R_1 \int \Theta_0^i u'(c_1^i) di - \frac{\psi_0}{\beta \Gamma R^*} \frac{R_1 \epsilon_0}{\epsilon_1} \\
 & c_t^i + b_{t+1}^i = R_t b_t^i + (A_t - \epsilon_t P_t^{T*} \theta) N_t - z_t^i + (R_t - 1) \underline{b}, \quad b_{t+1}^i \geq 0 \\
 & u'(c_t^i) \geq \tilde{\beta} R_{t+1} \mathbb{E}_t u'(c_{t+1}^i), \quad b_{t+1}^i [u'(c_t^i) - \tilde{\beta} R_{t+1} \mathbb{E}_t u'(c_{t+1}^i)] = 0 \\
 & \int c_t^i di + \bar{z} + \zeta \epsilon_t^{1+\chi} = A_t N_t \\
 & \sum_{t=1}^{\infty} \frac{P_t^{T*} \theta N_t + \frac{1}{\Gamma} \left(1 - \frac{R_{t+1}}{R^*} \frac{\epsilon_t}{\epsilon_{t+1}} \right)^2 - \zeta \epsilon_t^\chi}{(R^*)^{t-1}} = R^* B_1^{G*} + R_1 \frac{\epsilon_0 F_1^*}{\epsilon_1}
 \end{aligned}$$

- Initial states: $\{b_1^i, z_1^i, \Theta_0^i\}_i, R_1, \epsilon_0, F_1^*, B_1^{G*}, \psi_0$
- With no lump-sum transfers: household budget constraints and transversality are implementability conditions for the Ramsey planner.
- Timeless-commitment problem with past promises encoded in Lagrange multipliers ([Marcet and Marimon 2019](#)).

Heterogeneous Agents: Additional Policy Outcomes



- Counterfactual policy: 25% attenuated net capital inflows on impact, employment fixed at optimal.